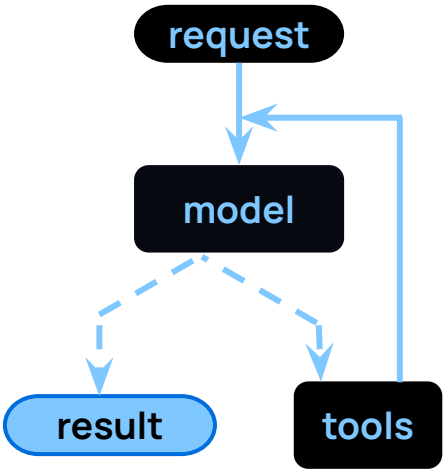
Loop Engineering with LangChain Deep Agents

Deep Agent Primer





create_agent

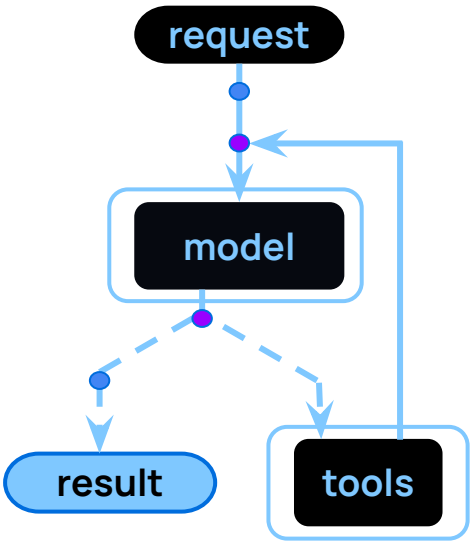



```python
from langchain.agents import create_agent

agent = create_agent(
    model="openai:gpt-5.5",
    tools=[get_weather],
    system_prompt="You are a helpful assistant",
)

result = agent.invoke(
    {"messages": [{"role": "user", "content":
"What's the weather in San Francisco?"}]}
)
```

middleware

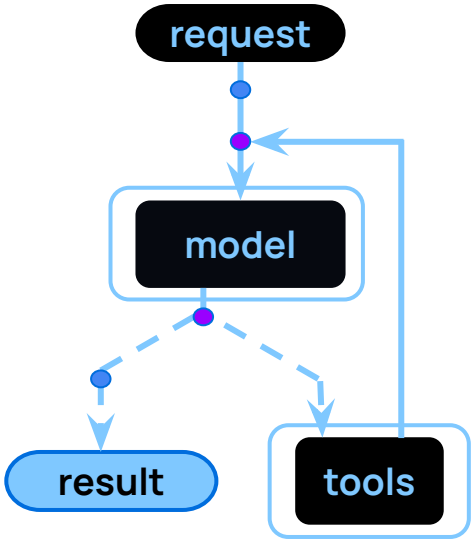


before_agent, after_agent

before_model, after_model

wrap_model_call, wrap_tool_call

create_deep_agent()



Execution Environment

- Tools
- Filesystem
- Code Interpreter
- Sandbox

Delegation

- Planning
- Subagents

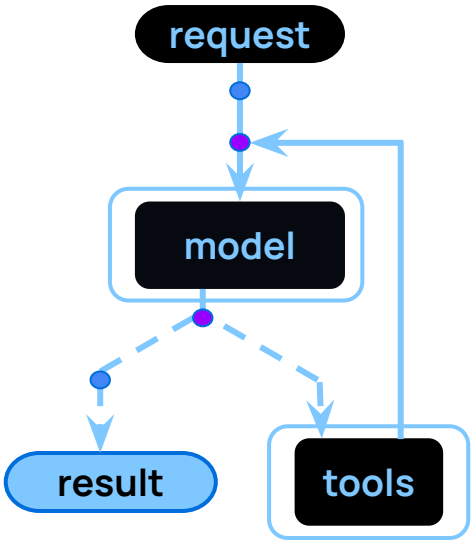
Context Management

- Memory: Agent.md
- Skills
- Summarization
- Context Offloading
- Prompt Caching

Steering

- Human-in-the-loop
- Grading Rubrics

create_deep_agent




```
backend = FilesystemBackend(root_dir="workspace", virtual_mode=True)

agent = create_deep_agent(
    model="openai:gpt-5.4-mini",
    tools=[fetch_requirement_from_jira],
    system_prompt="Follow the instructions in /memories/AGENTS.md.",
    memory=["/memories/AGENTS.md"],                # loaded on every run
    backend=backend,
    middleware=[ModelCallLimitMiddleware(run_limit=12, exit_behavior="end")],
)

result = agent.invoke({"messages": [{"role": "user", "content":
    "The requirement document is tracked in Jira as ticket AUTH-142. "
    "Fetch it to begin."}]})
```

Loop Engineering with LangChain Deep Agents

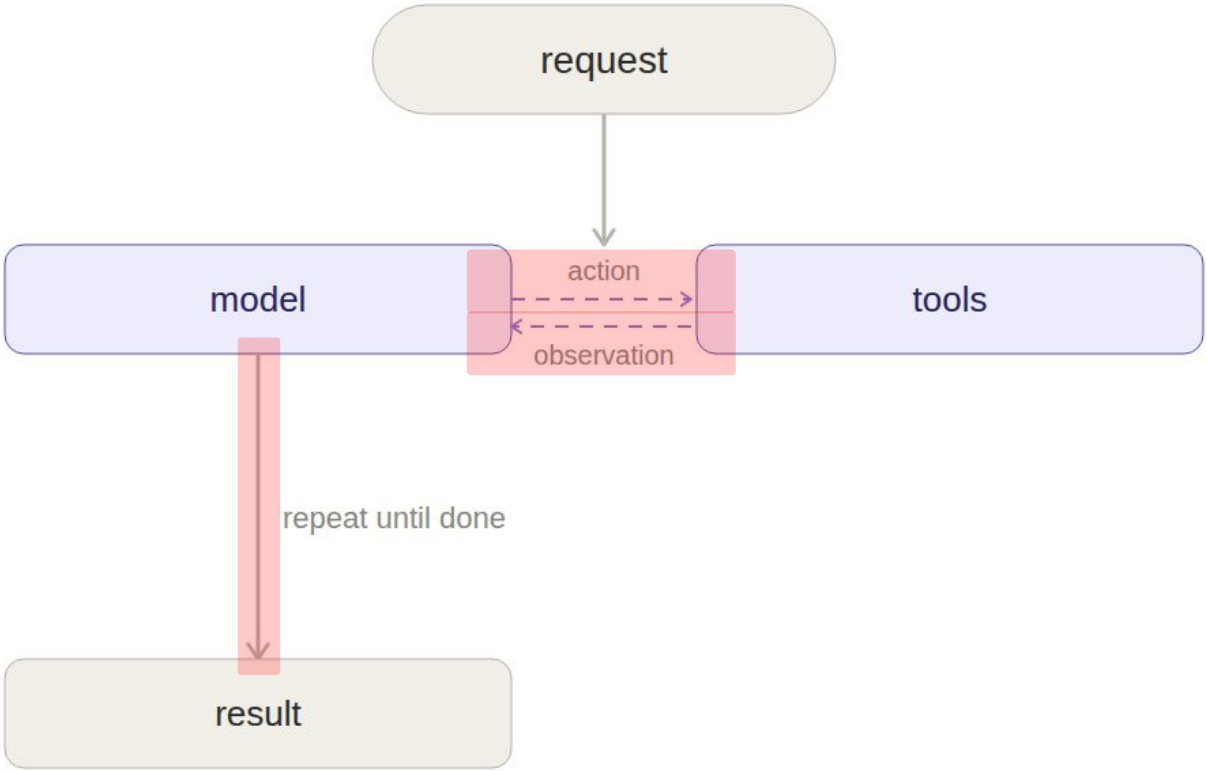
Agent Loop

LangChain

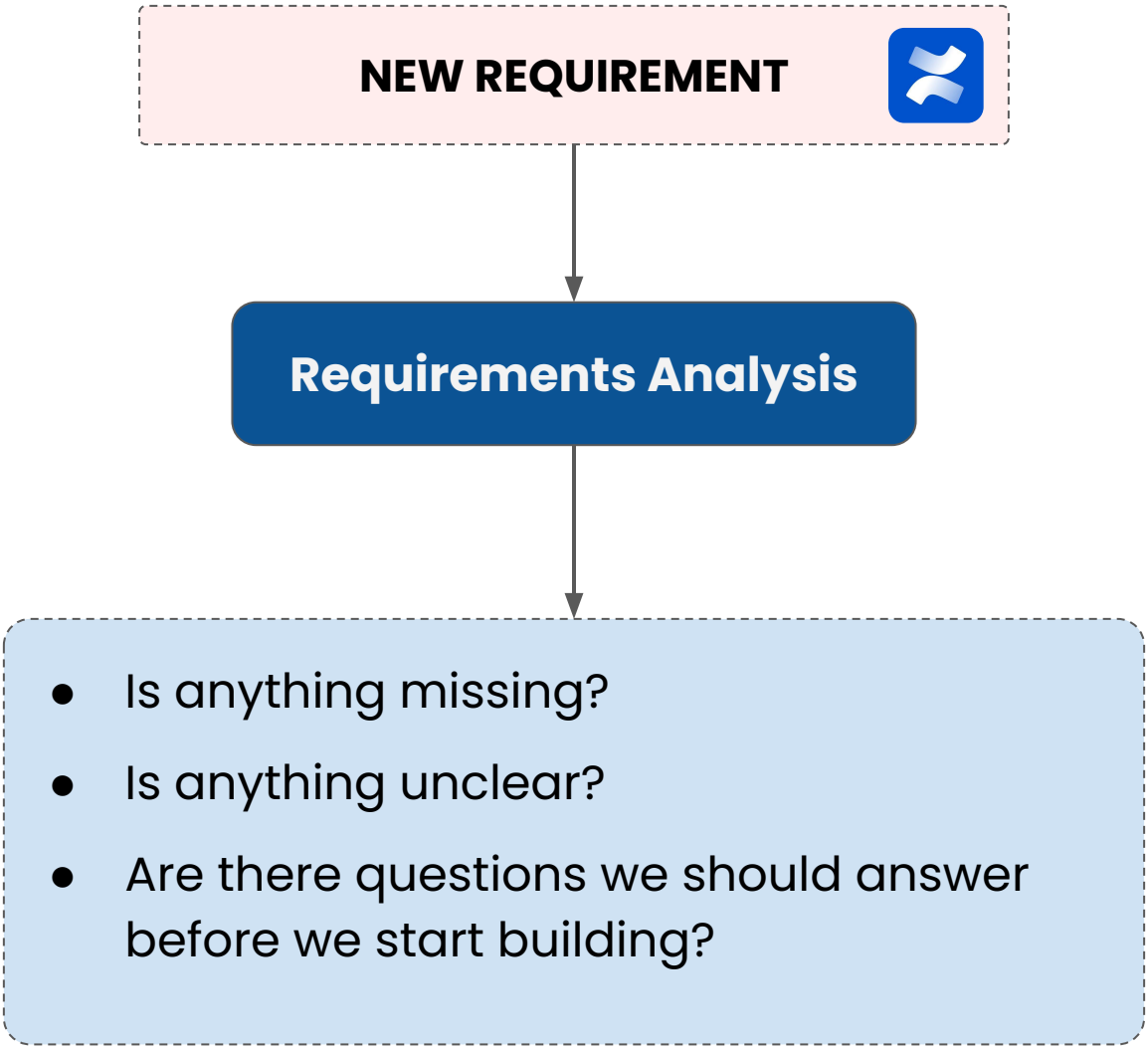


DeepLearning.AI

Agent Loop


request
model
action
observation
tools
repeat until done
result

Requirements Analysis Agent



Requirements Analysis Agent

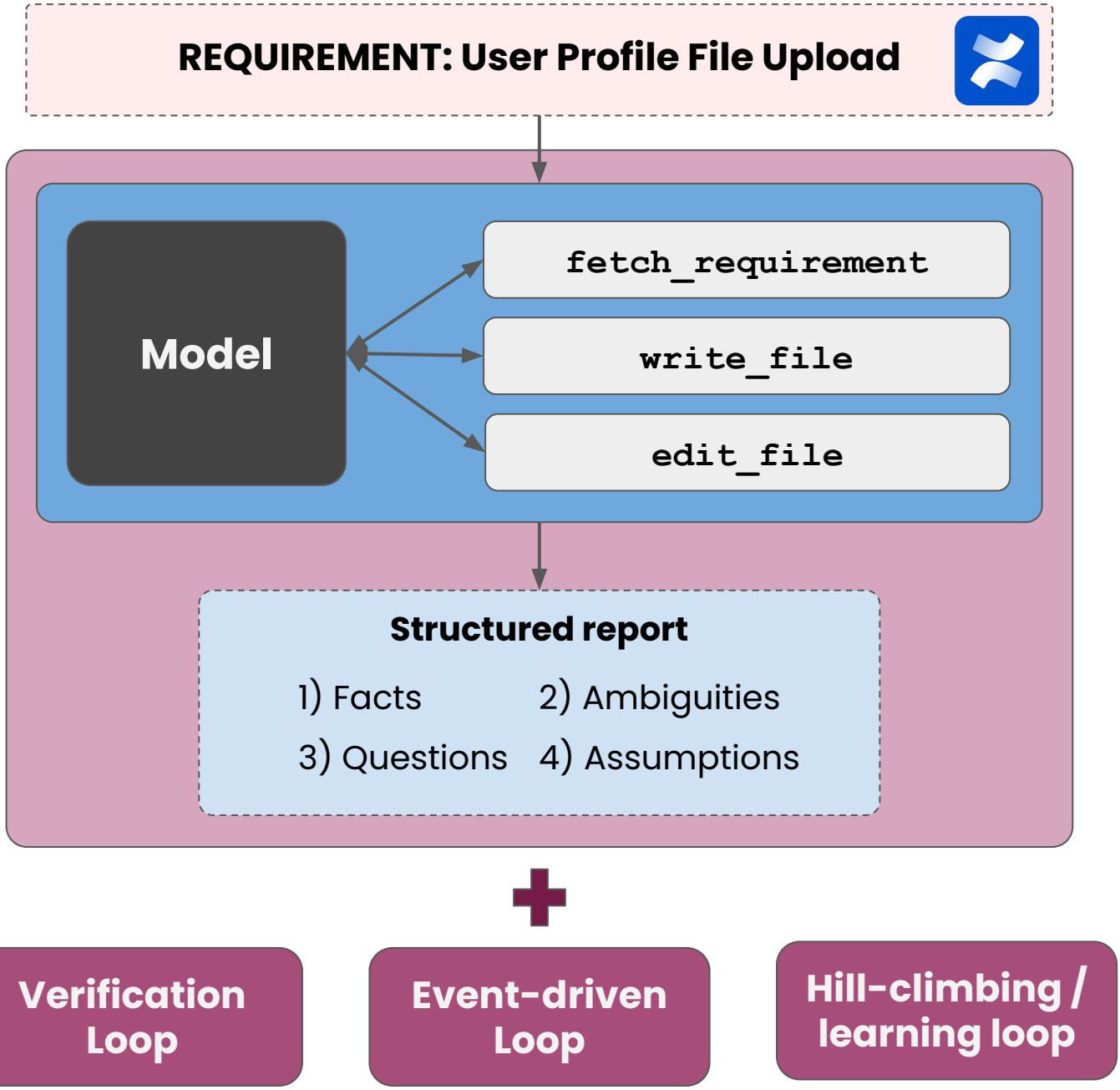
REQUIREMENT: User Profile File Upload

"Users shall be able to upload files to their profile. *Uploaded files should be stored* ***securely****. The system should support* ***common file types.***"

Requirements analysis agent

1. **Facts:** Users can upload files to their profile.
2. **Ambiguities:** "Common file types" is undefined. "Stored securely" doesn't specify a security standard.
3. **Clarification Questions:** What does "secure" require? Which file types must be supported (PDF, JPG, PNG, ...)?
4. **Assumptions:** Assume a 10MB max file size and encryption at rest, until confirmed otherwise.

Agent Tools


REQUIREMENT: User Profile File Upload
Model
fetch_requirement
write_file
edit_file
Structured report
1) Facts
2) Ambiguities
3) Questions
4) Assumptions
Verification Loop
Event-driven Loop
Hill-climbing / learning loop

Loop Engineering with LangChain Deep Agents

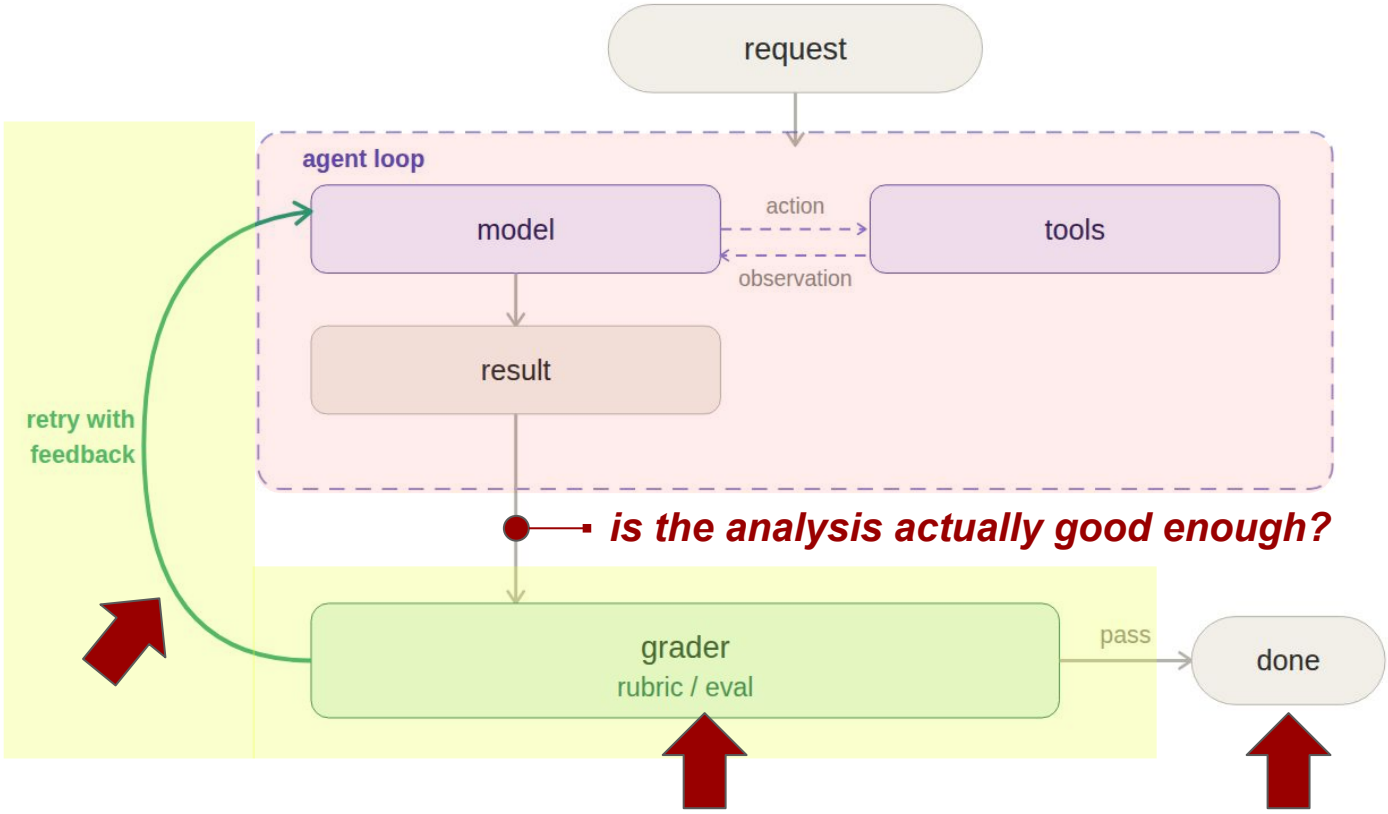
Verification Loop



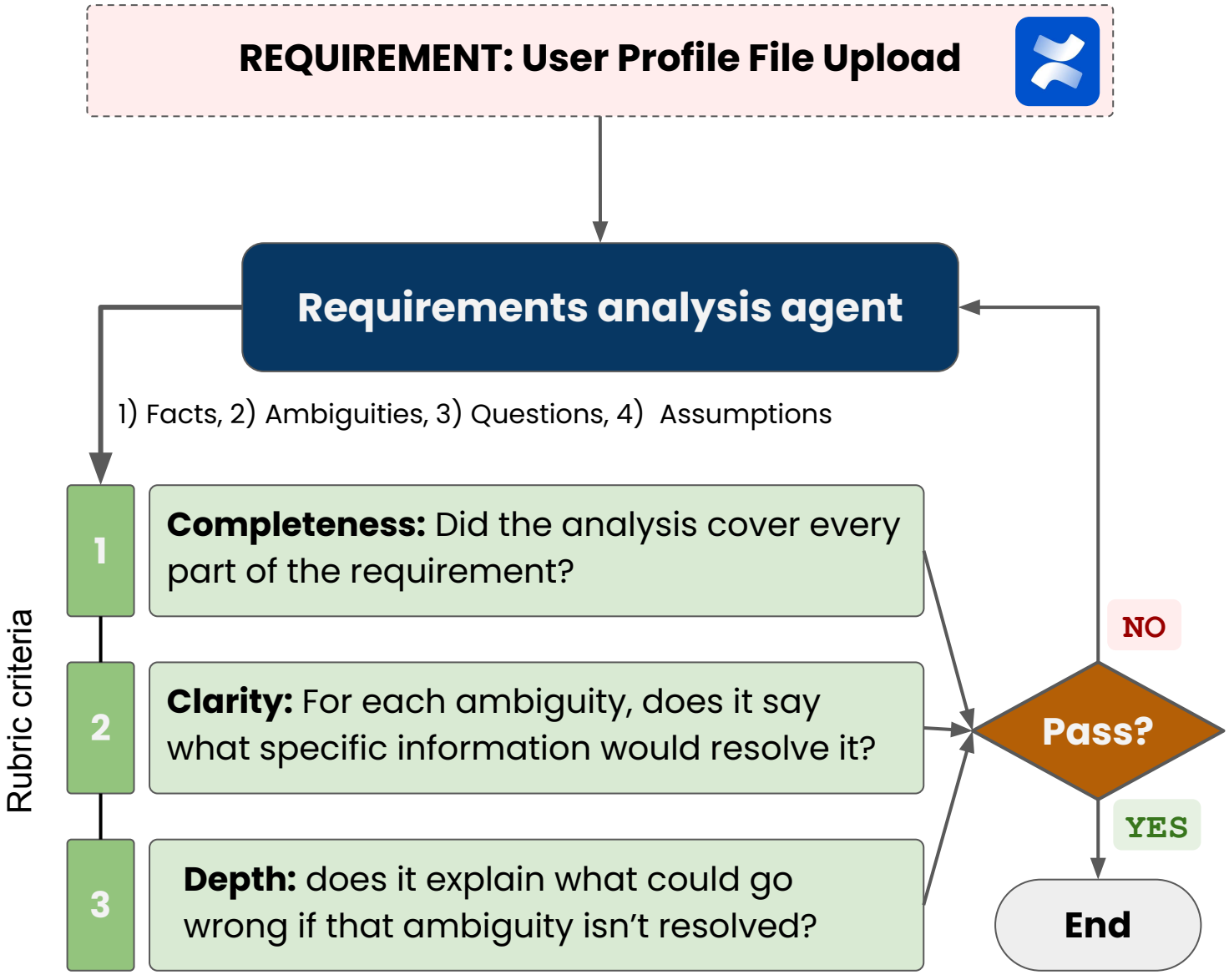


Verification Loop

Verification grades the output of a run of your agent and can retry the run with added feedback.



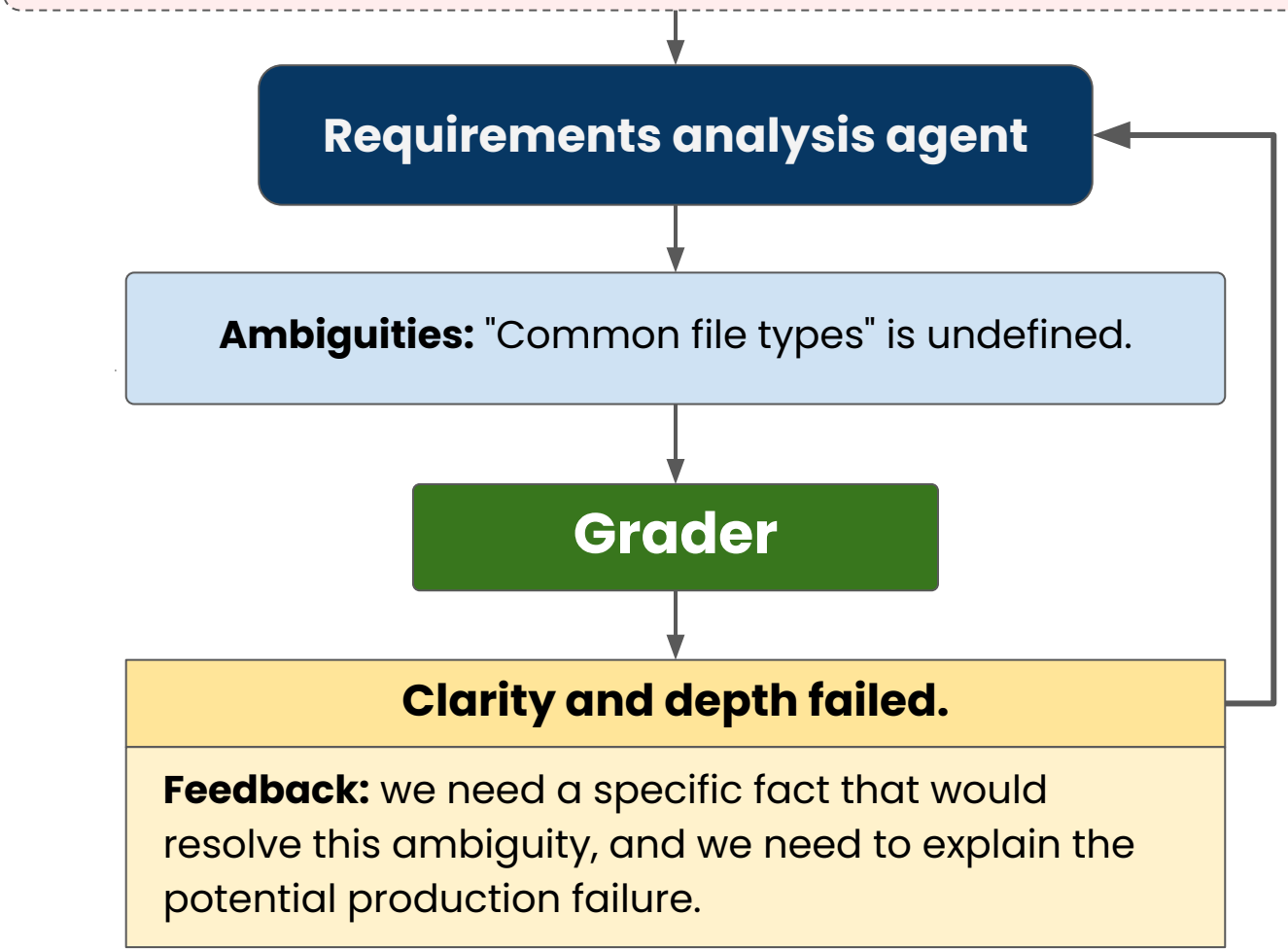
Agent Tools


REQUIREMENT: User Profile File Upload
Requirements analysis agent
1) Facts, 2) Ambiguities, 3) Questions, 4) Assumptions
Rubric criteria
1
Completeness: Did the analysis cover every part of the requirement?
2
Clarity: For each ambiguity, does it say what specific information would resolve it?
3
Depth: does it explain what could go wrong if that ambiguity isn't resolved?
Pass?
NO
YES
End

Requirements Analysis Agent

REQUIREMENT: User Profile File Upload

"Users shall be able to upload files to their profile. *Uploaded files should be stored* ***securely****. The system should support* ***common file types.***"



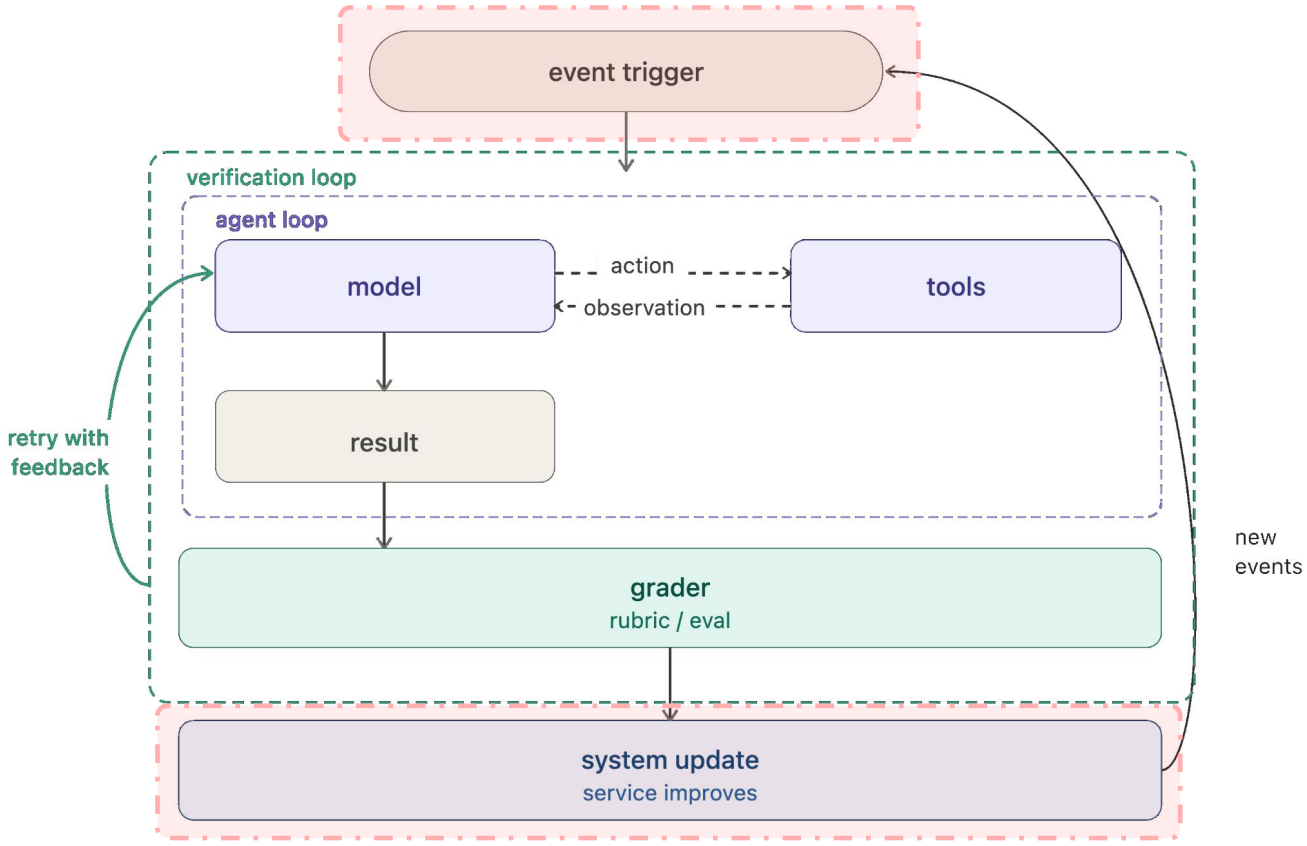
Loop Engineering with LangChain Deep Agents

Event driven loop

LangChain



DeepLearning.AI


event trigger
verification loop
agent loop
model
action
observation
tools
result
retry with feedback
grader
rubric / eval
new events
system update
service improves

Loop Engineering with LangChain Deep Agents

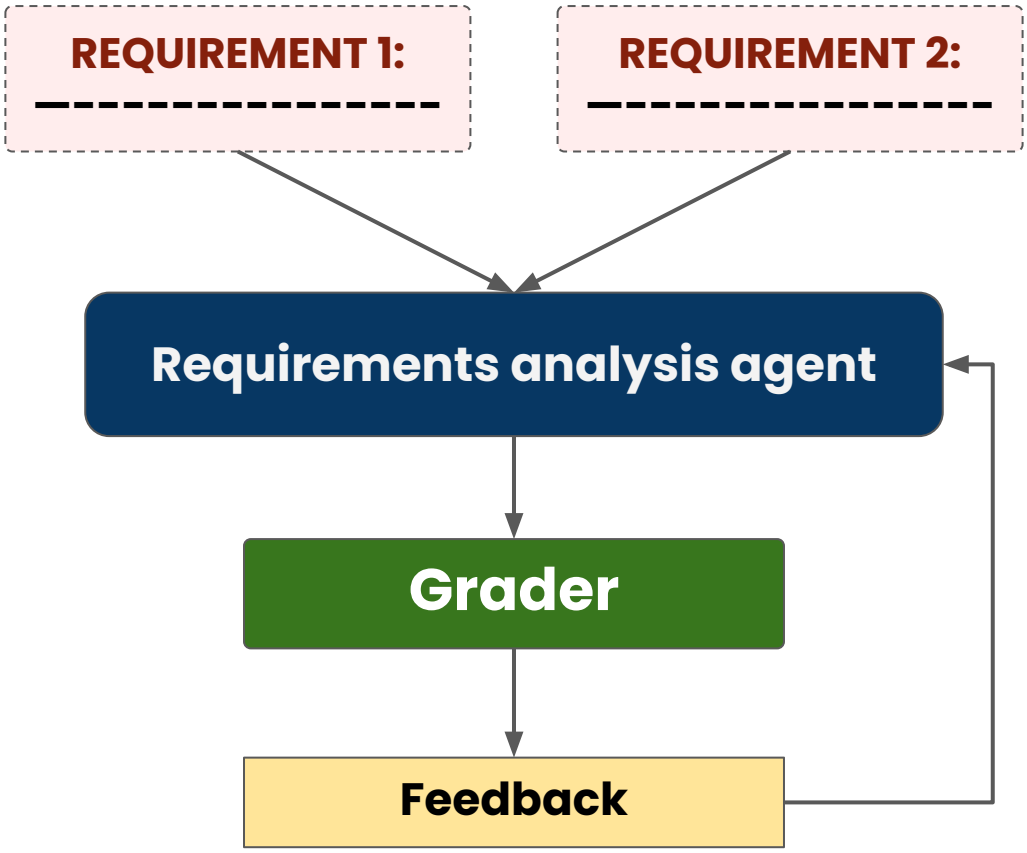
Learning loop

LangChain



DeepLearning.AI

Learning Loop



- Look across many runs
- Find recurring corrections
- Improve the agent for future runs